Helwan National University

Faculty of Engineering

Robotics & Mechatronics Department

ENGINEERING PROJECT REPORT

Uninterruptible Power Supply (UPS)

Microcontroller-Based Battery Backup System with SPWM Inverter

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Table of Contents

1. Introduction **1**

2. Objective **2**

3. UPS Definition **3**

3.1 Offline (Standby) UPS 3

3.2 Line-Interactive UPS 3

3.3 Online (Double-Conversion) UPS 4

4. Block Diagram **4**

4.1 Overall System Block Diagram 4

4.2 Control Path Block Diagram 5

5. Components Used **6**

5.1 Component Summary Table 6

5.2 Arduino Nano 7

5.3 Battery Bank 8

5.4 IRFZ44N Power MOSFETs 9

5.5 Center-Tapped Transformer 10

5.6 8-Pin Relay 12

5.7 2N2222 NPN Transistors 13

5.8 12 W LED Lamp (Load) 14

5.9 AC / DC Adapter and TP4056 Charging Module 14

5.10 Capacitors 16

5.11 AC Plug and Mains Input 17

6. Circuit Description **18**

6.1 Power Path 18

6.1.1 Normal (Mains) Power Path 18

6.1.2 Backup (Inverter) Power Path 18

6.2 Control Path 19
6.3 Power-Supply Rail Distribution 20
7. Arduino Control System 20
7.1 Pin Assignment and Signal Inversion 21
7.2 Timer1 Configuration and 20 kHz Switching Frequency 21
7.3 SPWM Technique — Sinusoidal Pulse Width Modulation 21
7.4 Output Frequency Verification — 50 Hz 22
7.5 Half-Cycle Logic — Q1 and Q2 Alternation 22
7.6 Dead Time and Shoot-Through Risk 23
7.7 Control-Flow Summary 23
7.8 Complete Arduino Source Code Listing 24
8. Proteus Simulation 26
8.1 Simulated Circuit Overview 26
8.2 Components Used in the Simulation 26
8.3 Arduino Behaviour in Simulation 27
8.4 Simulation of Mains Failure 28
8.5 Simulation of Backup Operation 28
9. Hardware Implementation 29
9.1 Overall Physical Implementation 29
9.2 Component Placement and Wiring 30
9.3 Battery and Charging Implementation 30
9.4 Construction Quality and Observations 31
10. Results & Discussion 31
10.1 Normal Operation (Mains Available) 31
10.2 Backup Operation (Mains Failed) 32
10.3 Inverter Output Waveform 32
10.4 Regulator Heating 33
10.5 Switching Stability 33
10.6 Summary of Observed Results 33

11. Errors & Problems Encountered 34

11.1 Error 1 — Relay Short-Circuit / Switching Fault.. 34

11.2 Error 2 — Square-Wave AC Output 35

11.3 Error 3 — Regulator Overheating 36

11.4 Error 4 — Insufficient MOSFET Gate Drive... 37

11.5 Error 5 — Q1 and Q2 Simultaneous Conduction (Shoot-Through)...................... 38

11.6 Error 6 — LED Lamp Pulsing... 38

12. Solutions Applied 39

12.1 Solutions Applied During the Project... 40

12.2 Recommended Future Improvements.. 40

12.2.1 Dedicated MOSFET Gate Driver 41

12.2.2 Explicit Dead-Time Control 41

12.2.3 Proper Relay Interlocking... 41

12.2.4 Electrical Isolation Between Control and Power Stages 41

12.2.5 Flyback and Transient Protection 41

12.2.6 Reverse-Voltage Protection.. 42

12.2.7 Improved Output Filtering.. 42

12.2.8 Modified-Sine or Pure-Sine Inverter Topology 42

12.2.9 Improved Battery Management 42

12.2.10 Better Thermal Management 42

12.3 Summary... 42

13. Advantages & Limitations 43

13.1 Advantages 43

13.2 Limitations... 44

14. Applications of UPS... 45

14.1 General Applications of UPS Systems 45

14.2 Applicability of this Prototype 46

14.3 Safety Considerations... 46

15. Conclusion 47

16. References.. 48

Note: This Table of Contents is generated via field codes. To refresh page numbers after editing, right-click the TOC and select "Update Field" → "Update entire table".

1. Introduction

Modern electrical and electronic systems depend on a continuous, high-quality alternating current (AC) power supply for correct and reliable operation. In practice, however, the public mains supply is never perfectly stable. Voltage sags, momentary interruptions, brownouts, transient surges and complete outages occur regularly due to grid switching operations, weather events, distribution faults and load variations. For loads that cannot tolerate even a few cycles of power loss — such as microcontroller-based equipment, communication devices, computers, security systems and lighting installations — these disturbances translate directly into data loss, system resets, hardware stress or complete operational failure.

The Uninterruptible Power Supply (UPS) is the standard engineering solution to this problem. A UPS is a self-contained electromechanical system that includes a backup energy source, a power-conditioning stage, a switching stage and a control unit. When the utility mains supply falls outside acceptable limits, the UPS automatically transfers the load to its internal backup source, typically a battery, and continues to supply power for a duration sufficient to either ride through the disturbance or to allow a controlled shutdown of the protected equipment.

This project documents the design, simulation, hardware implementation and testing of a practical UPS prototype intended for educational and laboratory use. The system is built around an Arduino Nano microcontroller and uses a push–pull MOSFET inverter stage, a 2N2222 transistor-based gate-driver network, a center-tapped transformer, an 8-pin relay for source selection, a Li-ion battery bank, and a TP4056-based charging module. The load used for testing is a 12 W LED lamp. The complete prototype demonstrates the full energy-conversion and control chain — from DC battery storage through high-frequency switching, voltage transformation, AC waveform synthesis and automatic source transfer.

The report follows a structured engineering format. It presents the design objectives, the underlying theory of UPS operation, the block diagram of the system, the component-level circuit description, the Arduino control algorithm including the Sinusoidal Pulse Width Modulation (SPWM) technique, the Proteus simulation results, the physical hardware implementation, and finally the observed results, problems encountered, applied solutions and recommendations for future improvement. All technical claims made in the report are supported either by the uploaded circuit diagrams, the Proteus simulation screenshots, the Arduino source code, or by direct inspection of the physical prototype.

It should be noted from the outset that the project is an educational prototype rather than a commercial product. The output power capability is limited, the output waveform is not a pure sinusoid, and several protection features that are mandatory in commercial UPS units are either simplified or omitted. These limitations are discussed openly in the relevant sections, since recognising them is itself an essential part of the engineering learning process.

2. Objective

The primary objective of this project is to design, simulate and physically implement a working Uninterruptible Power Supply capable of automatically maintaining power to a 12 W LED lamp load when the 220 VAC utility mains supply is interrupted. The project is intended to translate the theoretical concepts of power electronics, control systems and embedded programming into a tangible, testable hardware platform.

The specific technical objectives of the project are summarised below:

- Design a battery-backed inverter stage capable of converting DC battery energy into an alternating waveform suitable for driving a small AC load.
- Implement an automatic source-selection mechanism that supplies the load directly from the utility mains during normal operation and switches to the inverter output only when the mains supply fails.
- Use an Arduino Nano microcontroller as the central control element, generating the switching signals required to drive the MOSFET power stage.
- Apply a Sinusoidal Pulse Width Modulation (SPWM) technique at the Arduino output so that the inverter produces a waveform that approximates a sinusoidal envelope rather than a simple square wave.
- Verify the complete circuit in Proteus simulation software before physical construction, so that component-level behaviour can be inspected without risk to hardware.
- Construct a physical prototype on a breadboard using real components — including the Arduino Nano, IRFZ44N MOSFETs, 2N2222 transistors, a center-tapped transformer, an 8-pin relay and a Li-ion battery bank — and demonstrate its end-to-end operation.
- Identify, document and analyse the engineering problems encountered during testing, including relay switching faults, output waveform distortion, regulator heating and LED lamp instability, and propose technically sound solutions.

The scope of the project is intentionally limited to low-power demonstration. The prototype is not designed to replace a commercial UPS, nor is it intended to power high-wattage or sensitive electronic equipment. Instead, its purpose is to demonstrate, in a measurable and observable form, every functional block that a full-scale UPS contains:

energy storage, DC-to-AC conversion, automatic source selection and microcontroller-based control.

3. UPS Definition

An Uninterruptible Power Supply (UPS) is an electrical apparatus that provides emergency power to a load when the input power source — typically the utility mains — fails or becomes unacceptable in terms of voltage, frequency or waveform quality. Unlike an auxiliary or standby generator, a UPS is able to provide near-instantaneous power protection, typically within a few milliseconds, by drawing energy from storage devices such as sealed lead-acid batteries, lithium-ion cells or large capacitors.

The runtime of most UPS units is relatively short, usually a few minutes, but this is sufficient to start a secondary power source, to shut down protected equipment in a controlled manner, or to ride through short disturbances. The construction of a UPS therefore combines three essential engineering elements: an energy storage medium, a power-conditioning and conversion stage, and a control unit that decides when and how the stored energy should be delivered to the load.

From an architectural point of view, commercial UPS systems are generally classified into three main topologies:

3.1 Offline (Standby) UPS

In the offline or standby topology, the load is normally supplied directly from the utility mains through a transfer switch. The battery and inverter are kept idle, and only engage when the mains supply fails. The transfer time is typically a few milliseconds. This topology is simple, efficient and inexpensive, and is widely used for protecting single personal computers and small office equipment. The prototype described in this report belongs conceptually to this category, because the relay selects between the mains and the inverter output and the inverter only powers the load during backup operation.

3.2 Line-Interactive UPS

The line-interactive topology adds an automatic voltage regulation (AVR) stage between the mains input and the load. A bi-directional converter is connected in parallel with the load and is able to either buck or boost the mains voltage to compensate for moderate voltage variations without engaging the battery. This topology offers better protection against voltage sags and surges than the offline type, while still being more affordable than an online UPS.

3.3 Online (Double-Conversion) UPS

In the online or double-conversion topology, the load is always supplied from the inverter output. The incoming AC mains is first rectified to DC, which charges the battery and feeds the inverter; the inverter then synthesises a clean AC waveform for the load. There is no transfer switch in the conventional sense, because the inverter is always in the power path. This topology provides the highest power quality and zero transfer time, but is the most expensive and least efficient of the three.

The prototype implemented in this project is conceptually an offline (standby) UPS, since the load is normally fed from the mains through a relay and is switched to the inverter output only when the mains fails. This architecture was chosen because it best matches the educational objective of clearly separating the two operating modes (normal and backup) and because it requires fewer power-handling components than the online topology.

4. Block Diagram

The complete UPS system can be represented as a small number of functional blocks connected by two distinct signal paths: a power path, which carries the energy that ultimately reaches the load, and a control path, which carries the lower-power signals used to sense the state of the mains and to drive the inverter. Separating these two paths is essential for understanding the operation of the system and for diagnosing the problems encountered during testing.

4.1 Overall System Block Diagram

The following block diagram shows the principal functional blocks and the direction of energy flow through the system in both operating modes:

NORMAL PATH (mains available)

220 VAC Mains Input

↓

Relay — NO Contact

↓

Relay Common (COM)

↓

12 W LED Lamp Load

BACKUP PATH (mains failed)

Li-ion Battery Bank (~13.7 V)

↓

Push–Pull MOSFET Inverter (Q1, Q2)

↓

Center-Tapped Transformer (step-up)

↓

Relay — NC Contact

↓

Relay Common (COM) → Load

4.2 Control Path Block Diagram

The control path runs in parallel to the power path and is responsible for generating the switching signals that drive the inverter MOSFETs. It is built around the Arduino Nano and a 2N2222 transistor stage:

CONTROL PATH

Mains Sensing (relay coil driver)

↓

Relay State (NO / NC selection)

↓

Arduino Nano — SPWM Generator

↓

2N2222 Transistor Driver Stage (Q3, Q4)

↓

IRFZ44N MOSFET Gates (Q1, Q2)

↓

Center-Tapped Transformer Primary

In the implemented design, the relay coil is energised directly by the presence of the utility mains through a small rectifier/filter network. When the mains supply is present, the relay coil is energised and the relay common (COM) terminal is connected to the normally-open (NO) contact, routing mains voltage directly to the load. When the mains supply fails, the relay coil is de-energised and the COM terminal falls back to the normally-closed (NC) contact, which is fed by the inverter output transformer. This is a fail-safe arrangement: the load is automatically connected to the backup source in the absence of mains, even if the Arduino itself were to fail.

It should be noted that in the current implementation the Arduino continuously generates the SPWM signals that drive the MOSFET inverter, regardless of whether the mains is present or not. The Arduino does not therefore make an explicit decision to start the inverter on mains failure; instead, the inverter is always active and the relay simply decides which source (mains or inverter) is connected to the load at any given instant. This is a valid simplification for an educational prototype, but a more refined design would idle the inverter when the mains is healthy in order to reduce losses and component stress.

5. Components Used

The UPS prototype is assembled from a combination of low-voltage control components and higher-power switching components. The following table summarises the principal components used, together with their key specifications and functions. Detailed descriptions of each component follow the table.

5.1 Component Summary Table

Component	Quantity	Specification	Function
Arduino Nano	1	ATmega328P @ 16 MHz	Main control unit; generates SPWM signals
IRFZ44N MOSFET	2	N-channel, 55 V, 49 A, TO-220	Push–pull inverter switching
2N2222 NPN Transistor	2	NPN BJT, TO-92	MOSFET gate-driver stage (inverting)
Center-tapped Transformer	1	1 A (per spec.), 220 V ↔ low-V	Step-up (backup) / step-down (charging)
8-pin DPDT Relay	1	10 A 250 VAC, 12 V DC coil	Source selection (mains vs inverter)

Component	Quantity	Specification	Function
Li-ion Battery Bank	1 pack	~13.7 V nominal	Backup energy storage
TP4056 Charging Module	1	USB-C, with DW01 protection	Battery charging and protection
7805 Voltage Regulator	1	+5 V linear regulator	Powers Arduino Nano 5 V rail
AC/DC Adapter	1	Type-C Europlug → barrel jack	Alternate charging input (5 V)
Electrolytic Capacitor	1+	470 µF / 25 V	DC bus smoothing
Electrolytic Capacitor	1+	220 µF	Secondary rail filtering
Capacitor (ceramic / film)	1+	1 µF	High-frequency decoupling
Ceramic Capacitors (100 nF)	2	100 nF	7805 input/output decoupling
12 W LED Lamp	1	12 W, E27 base	Test load
AC Plug	1	220 VAC mains plug	Utility mains input
Status LED	1	Blue	Operational status indicator
Resistors (assorted)	—	10 Ω, 1 kΩ	Gate pull-up/down and base limiting

5.2 Arduino Nano

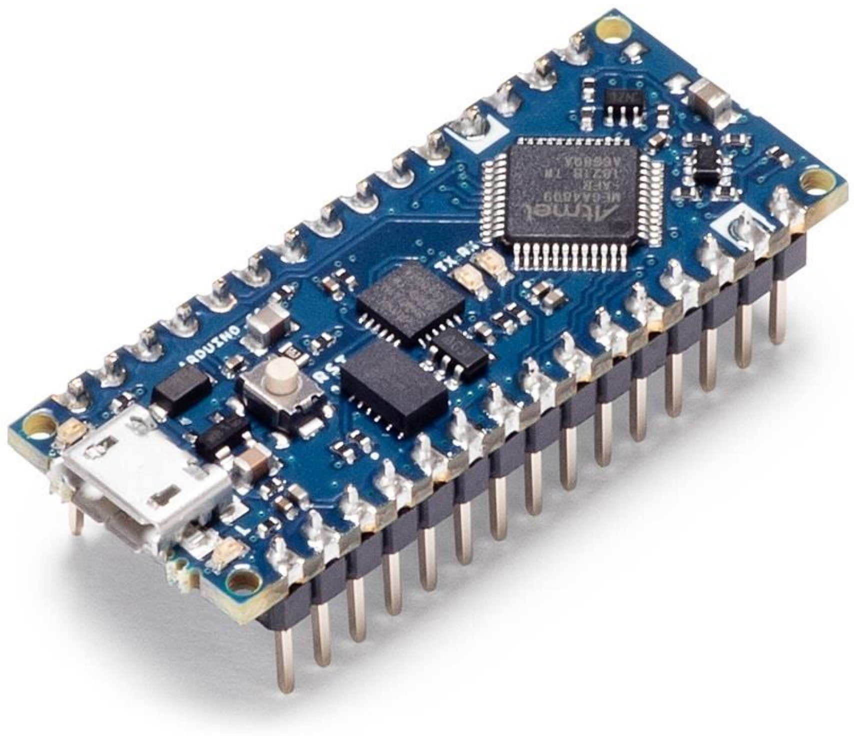

Figure 1. Arduino Nano microcontroller board. The ATmega328P microcontroller running at 16 MHz is visible at the centre of the board.

The Arduino Nano is a compact, breadboard-mountable microcontroller board based on the Atmel ATmega328P, running at a clock frequency of 16 MHz. It provides 14 digital input/output pins, 8 analog inputs, 6 PWM outputs and a serial UART interface. In this project the Arduino Nano acts as the central control unit of the UPS, generating the high-frequency switching signals that drive the MOSFET inverter stage.

Specifically, two of the Arduino's hardware PWM outputs (Timer1 channel A on pin D9 and channel B on pin D10) are used to produce the complementary SPWM signals required by the push–pull inverter. A third digital output (pin D3) drives a blue status LED that indicates the operational state of the controller. The Arduino is powered from a regulated +5 V rail derived from the battery through a 7805 linear regulator, allowing it to operate independently of the USB port once the program has been uploaded.

5.3 Battery Bank

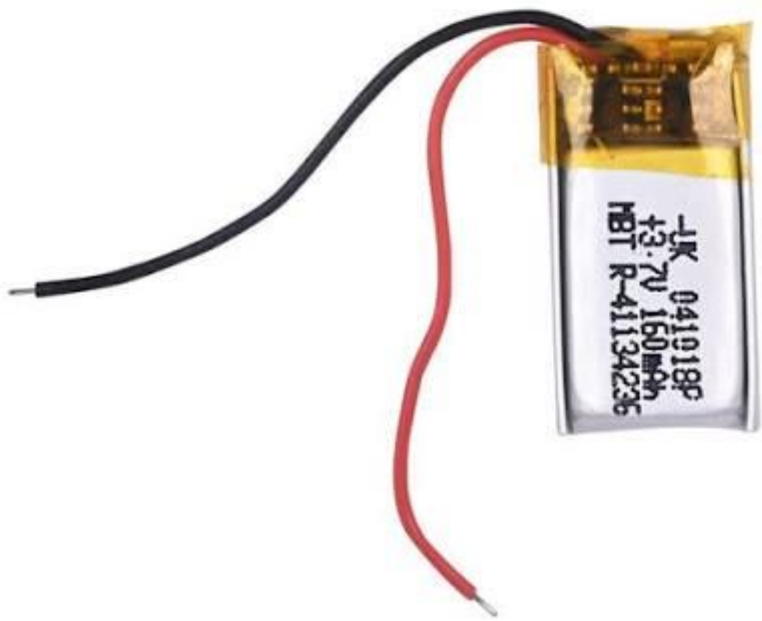

Figure 2. Lithium-polymer cell representative of the battery bank chemistry used in the project. The complete battery bank is rated at approximately 13.7 V nominal.

The backup energy source for the UPS is a battery bank with an approximate nominal voltage of 13.7 V, constructed from individual lithium-ion / lithium-polymer cells. The exact number of cells in the pack cannot be definitively determined from the available images, but the nominal voltage is consistent with a four-cell series configuration (4S, 4 × 3.45 V ≈ 13.8 V) or with a partially discharged state of a similar multi-cell pack. The battery bank provides the DC energy that is converted by the inverter stage into the AC waveform required by the load.

Battery charging is handled by a separate TP4056-based charging module (described below), which receives its input from a USB-C source and is responsible for the constant-current / constant-voltage charging profile required by lithium chemistry cells. The module also incorporates a DW01 protection IC and a dual 8205A MOSFET that protect the battery against overcharge, over-discharge and overcurrent conditions. The battery bank thus has both a charging path and a discharge path through the inverter.

5.4 IRFZ44N Power MOSFETs

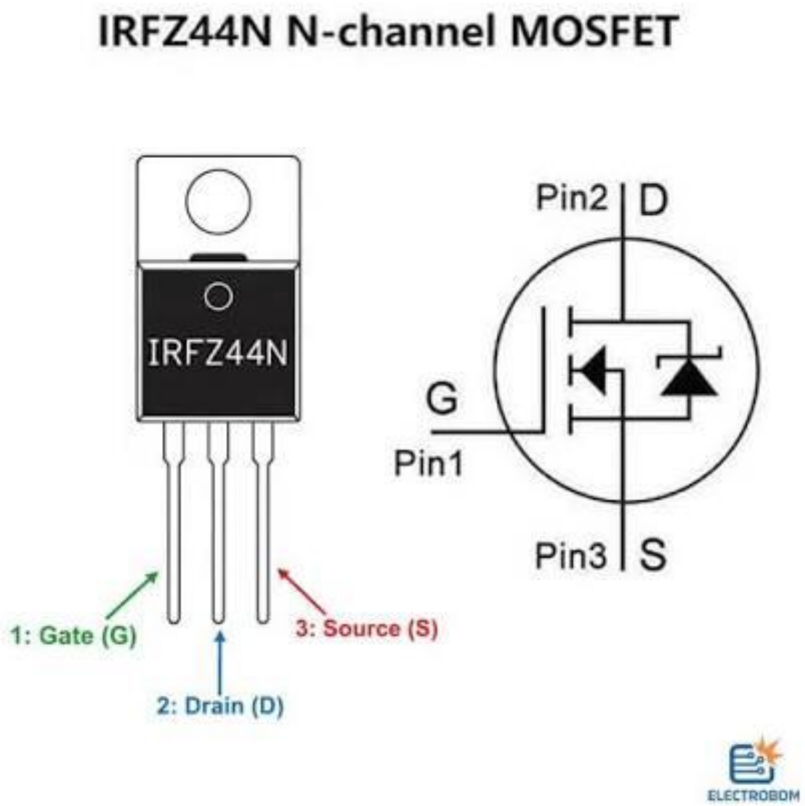

Figure 3. IRFZ44N N-channel power MOSFET in TO-220 package. Pinout (left to right): Gate, Drain, Source.

Two IRFZ44N N-channel enhancement-mode power MOSFETs are used as the main switching elements of the inverter stage. The IRFZ44N is rated for a drain-source voltage of 55 V and a continuous drain current of 49 A, with a low on-resistance, making it well suited for low-voltage push–pull inverter applications. Each MOSFET is responsible for switching one half-cycle of the AC output: one MOSFET drives current through one half of the center-tapped transformer primary, while the other MOSFET drives the opposite half during the alternate half-cycle.

The gates of the two MOSFETs are driven by a 2N2222 transistor stage rather than by a dedicated MOSFET gate-driver integrated circuit. This choice simplifies the circuit and reduces the component count, but, as discussed in the section on errors and problems, it has significant implications for switching performance, because the 2N2222 cannot source or sink the high instantaneous gate currents required for fast MOSFET switching.

5.5 Center-Tapped Transformer

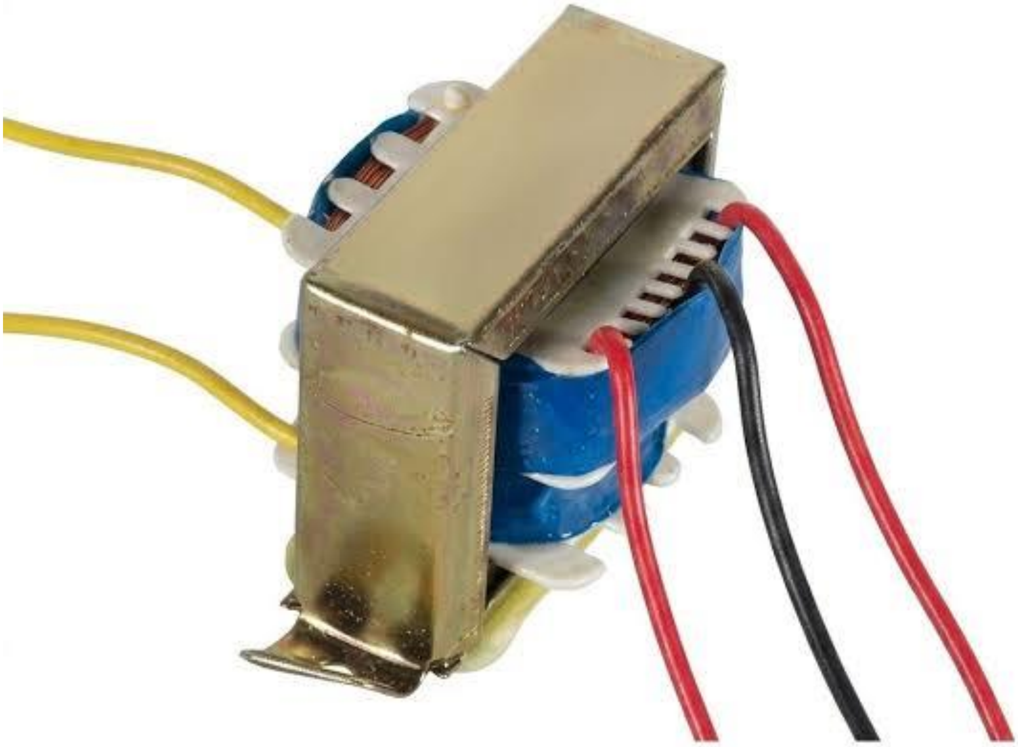

Figure 4. Center-tapped transformer used in the inverter stage. The primary winding has two terminals (yellow) and the secondary has three terminals (two red outer taps and one black center tap).

The transformer is a center-tapped unit with five wire leads: two on the primary side and three on the secondary side. The three secondary leads consist of two outer taps (red) and one center tap (black), which is the configuration required for a push–pull inverter. The transformer is used bidirectionally: in normal operation it acts as a step-down transformer that reduces the 220 V mains voltage to a low-voltage AC suitable for charging the battery (through a rectifier and the TP4056 module), while in backup operation it acts as a step-up transformer that raises the low-voltage switched battery voltage back to an AC level capable of driving the load.

The user-specified current rating of the transformer is 1 A. The standalone image of the transformer does not show any legible ratings printed on the casing, but a transformer visible in the full hardware photo is labelled AC 220 V~50 Hz, DC 0-12 V 500 mA — this may either be a different transformer used at an earlier stage of the prototype or a step-down transformer dedicated to the charging path. The exact relationship between these two transformers cannot be fully resolved from the available images and is noted here as an observation rather than a confirmed design choice.

5.6 8-Pin Relay

Figure 5. 8-pin DPDT electromagnetic relay (Songle SL-2 series). Rated 10 A at 250 VAC. Used as the source selector between mains and inverter output.

The source-selection element is an 8-pin DPDT (Double-Pole Double-Throw) electromagnetic relay of the Songle SL-2 series. The contact rating is 10 A at 250 VAC, which is more than sufficient for the 12 W (approximately 52 mA at 230 VAC) test load. The coil voltage is 12 V DC. The relay's common (COM) terminal is connected to one terminal of the load, the normally-open (NO) contact is connected to the 220 V mains input, and the normally-closed (NC) contact is connected to the inverter output.

This arrangement is fail-safe: when the mains supply is present, the relay coil is energised and the load is connected to the mains through the NO contact. When the mains supply is lost, the coil is de-energised and the COM terminal falls back to the NC contact, which is fed by the inverter output. This behaviour ensures that the load is automatically transferred to the backup source even in the event of a complete control-system failure.

During testing, the relay experienced a short-circuit / switching fault condition. This is discussed in detail in the section Errors & Problems Encountered.

5.7 2N2222 NPN Transistors

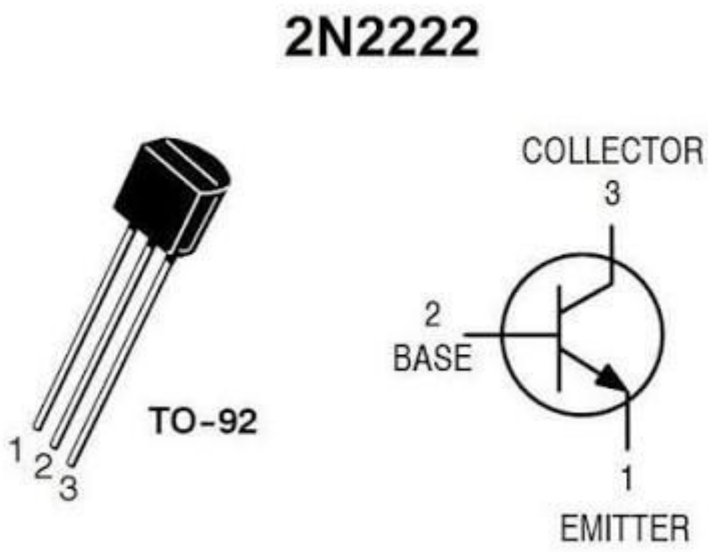

Figure 6. 2N2222 NPN transistor in TO-92 package. Pinout (left to right): Emitter, Base, Collector.

Two 2N2222 NPN bipolar junction transistors are used as the gate-driver stage between the Arduino digital outputs and the MOSFET gates. The 2N2222 is a general-purpose small-signal NPN switching transistor in a TO-92 package, capable of switching collector currents up to approximately 600–800 mA. In this circuit the transistors are configured as common-emitter inverting switches: when the Arduino output is HIGH, the transistor is turned ON and its collector pulls the corresponding MOSFET gate LOW, turning that MOSFET OFF. Conversely, when the Arduino output is LOW, the transistor is OFF and a pull-up resistor holds the gate HIGH, turning the MOSFET ON.

Because the transistor stage inverts the logic level of the Arduino output, the SPWM signal generated by the Arduino's Timer1 must also be configured in inverted mode (set via the COM1A0 and COM1B0 bits in the TCCR1A register). This double-inversion — once by the inverted PWM hardware, once by the transistor stage — restores the correct polarity at the MOSFET gate, so that a logical HIGH in the software SPWM waveform corresponds to the MOSFET being ON.

5.8 12 W LED Lamp (Load)

Figure 7. 12 W LED lamp used as the test load (SE HumMINGBIRD brand, E27 base). The lamp exhibited a pulsing behaviour during inverter operation, discussed in Section 11.

The test load is a 12 W LED lamp with an E27 Edison-screw base, manufactured under the brand SE HUMMINGBIRD. LED lamps are not simple resistive loads; they contain an internal electronic driver circuit that rectifies the incoming AC, smooths it to DC, and feeds a constant current to the LED string. This internal driver makes LED lamps particularly sensitive to the waveform quality of the supply: a pure sinusoidal AC waveform is handled without difficulty, but a distorted, square-wave or pulsed waveform can cause abnormal operation of the driver electronics, which may manifest as visible flicker, pulsing, partial illumination or complete failure to start.

As described in the section on errors and problems, the 12 W LED lamp used in this project did in fact exhibit a pulsing/flashing behaviour when supplied from the inverter output. This is consistent with the non-sinusoidal nature of the inverter waveform under the implemented gate-drive conditions, and is analysed in detail in the relevant sections.

5.9 AC / DC Adapter and TP4056 Charging Module

Figure 8. AC/DC wall adapter with Type-C Europlug input and barrel-jack DC output. Used as an alternate charging source for the battery.

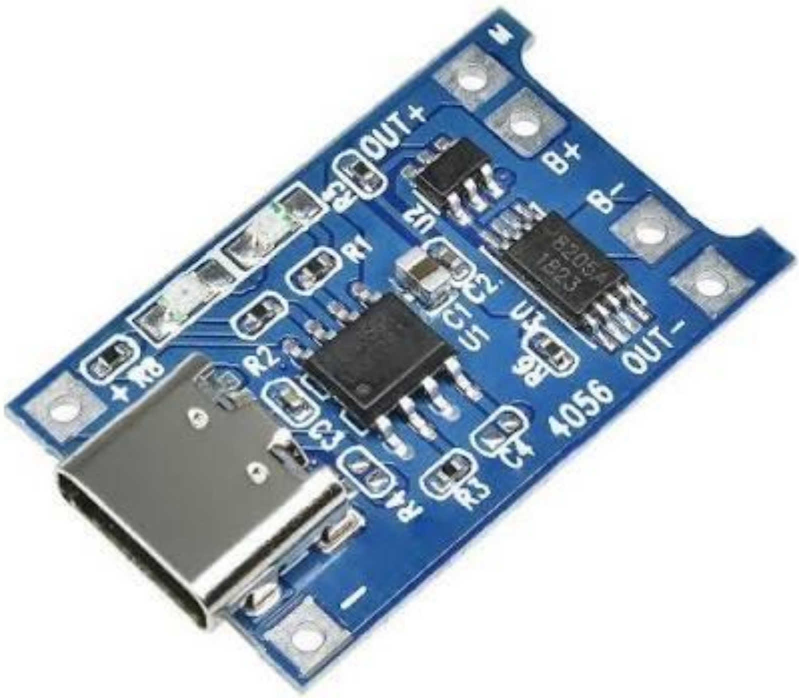

Figure 9. TP4056-based lithium battery charging module with USB-C input and integrated DW01 + 8205A battery protection circuit.

The battery is charged through a TP4056-based module that accepts a 5 V DC input via a USB-C connector and provides a regulated constant-current / constant-voltage charging profile for a single lithium cell. The module also integrates a DW01 battery-protection IC and a dual 8205A N-channel MOSFET, which together protect the battery against overcharge, over-discharge and overcurrent. The TP4056 module exposes three terminal pairs: IN+ / IN- (USB input), B+ / B- (battery), and OUT+ / OUT- (load output).

The 5 V input for the TP4056 can be provided either by a USB wall adapter or by an AC/DC adapter such as the one shown in the figure. The adapter accepts 220 V mains at its Type-C Europlug input and produces a low-voltage DC output at a barrel jack, suitable for feeding the USB-C input of the TP4056 module. This charging path is independent of the inverter stage and operates whenever the battery needs to be recharged.

5.10 Capacitors

Figure 10. 470 µF / 25 V aluminium electrolytic capacitor used for DC bus smoothing.

Three capacitor values are used in the circuit: 470 µF, 220 µF and 1 µF. The 470 µF / 25 V electrolytic capacitor is clearly visible in the component image and is used as a bulk smoothing capacitor across the main DC supply rail, reducing low-frequency ripple on the battery / regulator input. The 220 µF capacitor is used elsewhere on the breadboard for secondary rail filtering, and the 1 µF capacitor is used for high-frequency decoupling of noise on the supply or signal lines. Two additional 100 nF ceramic capacitors are used at the input and output of the 7805 linear regulator, as is standard practice.

The exact placement of every capacitor in the physical circuit could not be definitively resolved from the available images alone, since the breadboard wiring is dense and several capacitors are partially obscured by other components. The functional role of each capacitor is therefore described in terms of its general purpose (DC bus smoothing, regulator decoupling, noise filtering) rather than as a confirmed pin-accurate placement.

5.11 AC Plug and Mains Input

The 220 VAC utility mains is brought into the system through a standard AC plug. The plug feeds the relay's normally-open (NO) contact on one side and, in the charging path, the primary of the step-down transformer (or the AC input of the wall adapter) on the other.

Because the project handles live mains voltage, the safety implications are significant and are discussed in a dedicated subsection of the Safety discussion within the Applications section.

6. Circuit Description

The complete UPS circuit can be understood by tracing the two energy paths through the system: the main power path, which carries the energy delivered to the load, and the control path, which carries the low-power signals that drive the switching elements. This section describes each path in detail, based on the Proteus simulation and the physical hardware images.

6.1 Power Path

The power path consists of two parallel branches that converge at the relay's common (COM) terminal, which then feeds the load. The two branches are the mains (normal) branch and the inverter (backup) branch.

6.1.1 Normal (Mains) Power Path

When the 220 VAC utility mains is available, the power flows directly from the AC plug to the load through the relay's normally-open contact. The path is:

220 VAC Mains Plug

↓

Relay NO Contact

↓

Relay COM Terminal

↓

12 W LED Lamp

In this mode the relay coil is energised by the mains supply (through a small rectifier and filter network), and the COM terminal is held against the NO contact. The inverter continues to run, but its output is delivered to the relay's NC contact, which is not connected to the load, so no inverter power reaches the load during normal operation. The battery is simultaneously charged through the TP4056 module.

6.1.2 Backup (Inverter) Power Path

When the mains supply fails, the relay coil is de-energised and the COM terminal falls back to the NC contact, which is fed by the inverter output transformer. The backup power path is:

Li-ion Battery Bank (~13.7 V DC)

↓

Push–Pull MOSFET Stage (Q1, Q2)

↓

Center-Tapped Transformer Primary

↓

Transformer Secondary (stepped-up AC)

↓

Relay NC Contact

↓

Relay COM Terminal → Load

The MOSFETs Q1 and Q2 alternately switch the battery voltage through the two halves of the transformer's center-tapped primary winding. Because the two halves of the primary are switched in opposite phase, the resulting flux in the transformer core alternates, and an AC voltage is induced in the secondary winding. The turns ratio of the transformer is selected so that the secondary voltage is sufficient to drive the load when the primary is switched by the battery voltage.

6.2 Control Path

The control path is responsible for generating the switching signals that drive the MOSFET gates. It begins at the Arduino Nano and ends at the MOSFET gates, passing through the 2N2222 transistor driver stage. The path is:

Arduino Nano — Timer1 SPWM Generator

↓

Base Resistors (1 kΩ) → 2N2222 Bases (Q3, Q4)

↓

2N2222 Common-Emitter Inverters

↓

Gate Pull-Up / Pull-Down Resistors (10 Ω)

↓

IRFZ44N MOSFET Gates (Q1, Q2)

The Arduino Nano generates two complementary SPWM signals on pins D9 and D10, using the Timer1 hardware in Fast PWM mode with ICR1 as the TOP value. Each signal corresponds to one half-cycle of the desired 50 Hz output: when one signal is actively generating the sinusoidal envelope, the other is held at zero. The 2N2222 transistors invert these signals (so that an Arduino HIGH turns the corresponding MOSFET OFF) and the inverted PWM mode of Timer1 compensates for this inversion, resulting in the correct gate-drive polarity.

6.3 Power-Supply Rail Distribution

In addition to the main power and control paths, the circuit contains a small power-distribution network that supplies the low-voltage control components. The battery voltage (approximately 13.7 V) feeds the VIN pin of the Arduino Nano and also feeds the input of a 7805 linear regulator. The 7805 produces a regulated +5 V output, which is decoupled by 100 nF ceramic capacitors at its input and output, and this regulated rail powers the Arduino's 5 V pin and the blue status LED. The 2N2222 transistors, the relay coil drive (when present), and the TP4056 charging module each have their own dedicated supply arrangements, all referenced to the common battery negative terminal that serves as the system ground.

It should be noted that the 7805 regulator became hot during testing, and this overheating was associated with reverse voltage coming from the lamp/output side of the circuit. This issue is discussed in detail in Section 11.

7. Arduino Control System

The Arduino Nano is the brain of the UPS. It continuously generates the high-frequency switching signals required by the push–pull MOSFET inverter, using a technique known as Sinusoidal Pulse Width Modulation (SPWM). This section explains in detail how the Arduino code works, why each configuration choice was made, and how the resulting signals drive the power stage.

The control program is written in C++ using the Arduino integrated development environment (Arduino IDE), which uses the AVR-GCC toolchain underneath. The complete listing of the program is presented later in this section; the following subsections explain its operation step by step.

7.1 Pin Assignment and Signal Inversion

The code defines three output pins: Q1 on pin D9, Q2 on pin D10 and a status LED on pin D3. Pins D9 and D10 are deliberately chosen because they correspond to the Output Compare A (OC1A) and Output Compare B (OC1B) channels of the Arduino's 16-bit Timer1, which is capable of generating hardware PWM with a user-defined TOP value. This hardware PWM is essential for producing a clean, jitter-free high-frequency switching signal.

A key consideration is the relationship between the Arduino output and the MOSFET gate. Because the 2N2222 transistor stage is wired as a common-emitter inverter, an Arduino HIGH output turns the 2N2222 ON, which pulls the MOSFET gate LOW, which in turn switches the MOSFET OFF. The signal therefore experiences one logical inversion between the Arduino and the MOSFET gate. To compensate for this, the Timer1 PWM is configured in inverted mode by setting both the COM1A0 and COM1B0 bits in the TCCR1A register (in addition to the COM1A1 and COM1B1 bits). The double inversion — once in software (inverted PWM mode) and once in hardware (transistor inverter) — restores the intended polarity at the MOSFET gate.

7.2 Timer1 Configuration and 20 kHz Switching Frequency

Timer1 is configured in Fast PWM mode with ICR1 as the TOP value. This corresponds to Waveform Generation Mode 14 (WGM13:0 = 1110), which is selected by setting the WGM11, WGM13 and WGM12 bits in TCCR1A and TCCR1B. The prescaler is set to 1 (CS10 bit only), so the timer is clocked directly from the Arduino's 16 MHz system clock without division.

With ICR1 set to 799, the timer counts from 0 to 799 (a total of 800 counts) before resetting. The resulting PWM carrier frequency is:

$$f_{PWM} = f_{CPU} / (N \times (1 + TOP)) = 16\text{ MHz} / (1 \times 800) = 20\text{ kHz}$$

A 20 kHz switching frequency is well above the audible range and is a common choice for SPWM inverters. At this frequency, each MOSFET switches approximately 20,000 times per second, and the resulting high-frequency PWM can be filtered down to a low-frequency sinusoidal envelope by the transformer and any additional filter components.

7.3 SPWM Technique — Sinusoidal Pulse Width Modulation

Sinusoidal Pulse Width Modulation (SPWM) is the technique used to synthesise a low-frequency sinusoidal waveform from a high-frequency PWM carrier. The Arduino does not output a pure sine wave directly; instead, it outputs a high-frequency PWM signal whose duty cycle is varied in accordance with a sinusoidal reference. When this PWM signal is averaged by a low-pass filter (or by the inductance of the transformer itself), the result is a waveform whose envelope approximates a sine wave.

In this implementation, the duty cycle within each half-cycle is generated by sampling a sine curve at 50 discrete points. The duty cycle value written to the Output Compare Register (OCR1A or OCR1B) at each sample is:

$$\mathbf{OCR1x = 799 \times \sin(\pi \times i / 49), \quad for\ i = 0 \ldots 49}$$

At i = 0, the sine value is zero, so the duty cycle is 0 %. At i = 24 (the middle of the half-cycle), the sine value is sin(π/2) = 1, so the duty cycle reaches its maximum of 799/799 = 100 %. At i = 49, the sine value returns to sin(π) = 0, so the duty cycle returns to 0 %. The duty-cycle envelope therefore traces a half-sine over 50 samples, which is exactly the shape required for one half-cycle of the AC output.

7.4 Output Frequency Verification — 50 Hz

The 50 samples of each half-cycle are spaced by a delay of 200 microseconds, so the duration of one half-cycle is:

$$\mathbf{T\_half = 50 \times 200\ \mu s = 10\ ms}$$

A full cycle consists of two half-cycles (one driven by Q1, the next by Q2), so the full-cycle duration is 20 ms. The output frequency is therefore:

$$\mathbf{f\_out = 1 / T = 1 / 20\ ms = 50\ Hz}$$

This is exactly the standard European / African mains frequency of 50 Hz, which means the SPWM envelope synthesised by the Arduino is correctly matched to the frequency that the lamp load and the transformer are designed to operate at.

It is important to distinguish between the two frequencies present in the system. The 20 kHz carrier is the rate at which the MOSFETs actually switch, and is the frequency that is filtered out by the transformer and any additional low-pass network. The 50 Hz envelope is the desired output frequency at the load, and is the frequency at which the lamp operates. These two frequencies are independent and serve different purposes in the SPWM scheme.

7.5 Half-Cycle Logic — Q1 and Q2 Alternation

Each MOSFET is responsible for one half-cycle of the AC output. During the first half-cycle (positive half of the sine), Q1 is actively driven with the SPWM pattern while Q2 is held OFF (OCR1B = 0). During the second half-cycle (negative half of the sine), the roles are reversed: Q2 is actively driven with the SPWM pattern while Q1 is held OFF. This alternation is implemented by the two for-loops in the main loop() function.

This is the standard push–pull inverter drive scheme: only one MOSFET is conducting at any instant, and the two MOSFETs together cover the full sine period. The center-tapped transformer primary combines the two half-cycle currents into a single alternating flux in the core, which induces the AC output voltage on the secondary winding.

7.6 Dead Time and Shoot-Through Risk

An important consideration in any push–pull inverter is the so-called dead time — the small idle interval between the turn-OFF of one MOSFET and the turn-ON of the other, during which both devices are guaranteed to be OFF. Dead time is necessary because MOSFETs do not switch instantaneously; if the OFF-going MOSFET has not fully turned OFF before the ON-going MOSFET begins to turn ON, both devices may conduct simultaneously, creating a low-impedance path (shoot-through) directly across the battery. The resulting current spike can be very large and may destroy the MOSFETs.

In the current Arduino code, no explicit dead time is inserted between the two half-cycles. The code simply transitions from one for-loop to the other, setting OCR1A = 0 and OCR1B = 0 at the boundary. While the SPWM pattern naturally ramps the duty cycle down to zero at the end of each half-cycle (since sin(π) = 0), the gate-drive transition speed is governed by the 2N2222 transistor stage, which — as discussed in Section 11 — does not provide the strong, fast gate drive that would be required for clean, low-loss switching. The combination of these two factors creates a real risk of momentary shoot-through at the half-cycle boundaries, which is one of the principal weaknesses of the current implementation.

7.7 Control-Flow Summary

The complete control flow executed by the Arduino can be summarised as follows:

START — Power-on / Reset

↓

Initialise I/O pins (Q1, Q2, LED)

↓

Set Q1, Q2, LED HIGH (MOSFETs OFF initially)

↓

Configure Timer1: Fast PWM, ICR1 = 799, inverted

↓

OCR1A = 0, OCR1B = 0 (no output initially)

↓

LOOP: Generate Q1 half-sine (50 samples × 200 µs)

↓

Generate Q2 half-sine (50 samples × 200 µs)

↓

Repeat loop indefinitely

It should be noted that the current implementation does not include explicit mains-sensing logic in the Arduino firmware. The Arduino continuously runs the inverter regardless of the mains state, and the source-selection decision is delegated entirely to the relay, which is autonomously controlled by the presence or absence of the mains supply. This simplifies the control software considerably, but means that the inverter is always consuming battery power even when the mains is healthy — a limitation that would be addressed in a more refined design.

7.8 Complete Arduino Source Code Listing

The complete Arduino source code, as actually uploaded to the hardware, is listed below:

```
#include <math.h>

const int Q1 = 9;
const int Q2 = 10;
const int LED = 3;

void setup() {
  pinMode(Q1, OUTPUT);
  pinMode(Q2, OUTPUT);
  pinMode(LED, OUTPUT);
```

```
  // With 2N2222 inverting stage: Arduino HIGH = MOSFET OFF
  digitalWrite(Q1, HIGH);
  digitalWrite(Q2, HIGH);
  digitalWrite(LED, HIGH);

  // Timer1 — 20 kHz SPWM (inverted PWM due to 2N2222 inversion)
  TCCR1A = _BV(COM1A1) | _BV(COM1A0) |
           _BV(COM1B1) | _BV(COM1B0) |
           _BV(WGM11);

  TCCR1B = _BV(WGM13) | _BV(WGM12) | _BV(CS10);

  ICR1 = 799;
  OCR1A = 0;
  OCR1B = 0;
}

void loop() {
  // Q1 half-sine
  for (int i = 0; i < 50; i++) {
    OCR1A = 799 * sin(PI * i / 49.0);
    OCR1B = 0;
    delayMicroseconds(200);
  }
  // Q2 half-sine
  for (int i = 0; i < 50; i++) {
    OCR1A = 0;
    OCR1B = 799 * sin(PI * i / 49.0);
    delayMicroseconds(200);
  }
}
```

This code is compiled and uploaded to the Arduino Nano through the Arduino IDE. Once running, it continuously generates the SPWM pattern that drives the inverter MOSFETs, producing a 50 Hz AC waveform at the transformer secondary.

8. Proteus Simulation

Before constructing the physical prototype, the complete UPS circuit was modelled and verified in Proteus Design Suite, a widely used electronic design automation environment that combines schematic capture with mixed-signal simulation. Proteus is particularly well suited to this project because it includes a built-in model for the Arduino Nano, allowing the actual firmware (compiled from the Arduino IDE) to be executed in the simulator and to interact with the surrounding analogue and digital components in real time.

8.1 Simulated Circuit Overview

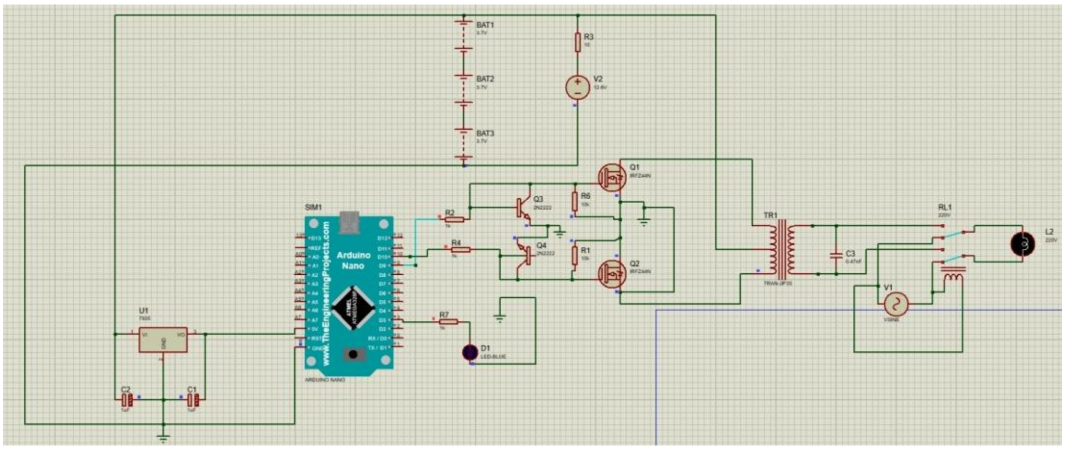

Figure 11. Full Proteus simulation schematic of the UPS circuit. The Arduino Nano (centre-left) drives two 2N2222 transistors, which in turn drive two IRFZ44N MOSFETs in a push–pull configuration with a center-tapped transformer and an 8-pin relay.

The Proteus schematic includes all the principal components of the physical prototype: an Arduino Nano microcontroller (designated U2), two IRFZ44N power MOSFETs (Q1 and Q2), two 2N2222 NPN driver transistors (Q3 and Q4), a 7805 +5 V linear regulator (U1), a center-tapped transformer (TR1), an 8-pin relay (RL1), a multi-cell battery bank (BAT1, BAT2, BAT3), several decoupling and filter capacitors (C1, C2, C3), a set of base-limiting and pull-up / pull-down resistors, a blue status LED (D1) and the 230 V lamp load (L2). The schematic also includes two virtual instruments: a DC voltmeter (V2) displaying the battery rail voltage, and an AC voltmeter (V1) representing the mains input.

8.2 Components Used in the Simulation

The components placed in the Proteus schematic, together with their designators and key parameters, are summarised below:

Designator	Component	Value / Model	Role
U2	Arduino Nano	SIM1 (ATmega328P)	SPWM generation

Designator	Component	Value / Model	Role
Q1, Q2	N-channel MOSFET	IRFZ44N	Push–pull inverter switches
Q3, Q4	NPN BJT	2N2222	Inverting gate drivers
U1	Linear regulator	7805	+5 V rail for Arduino
TR1	Center-tapped transformer	TRAN-2P3S	DC-to-AC conversion / voltage step-up
RL1	Electromagnetic relay	8-pin, 300 V-rated	Source selection
BAT1–3	DC sources (battery model)	9.7 V each (sim abstraction)	Backup energy
C1, C2	Ceramic capacitor	100 nF	7805 decoupling
C3	Film capacitor	647 nF	Secondary filtering
R1, R3, R5, R6	Resistor	10 Ω	Gate pull-up / pull-down / current sense
R2, R4, R7	Resistor	1 kΩ	2N2222 base limiters
D1	Status LED	Blue	Operational indicator
L2	Lamp load	LAMP 230V	Test load
V1	AC voltmeter (virtual)	VSINE	Mains input simulation
V2	DC voltmeter (virtual)	14.8 V (reading)	Battery rail measurement

It should be noted that the battery representation in Proteus (three sources of 9.7 V each, summing nominally to 29.1 V) is a simulation abstraction that does not exactly match the physical 13.7 V battery bank. The DC voltmeter in the simulation reads 14.8 V at the Arduino VIN node, which is more representative of the physical hardware. This discrepancy is a known artefact of the simulation model and does not affect the validity of the functional verification.

8.3 Arduino Behaviour in Simulation

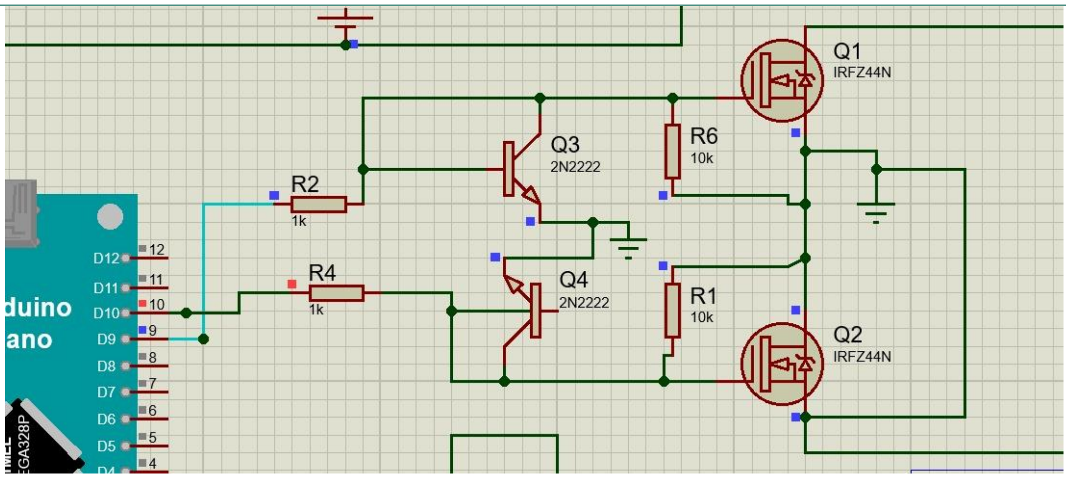

Figure 12. Proteus simulation — zoomed view of the Arduino Nano, the 2N2222 transistor drivers and the IRFZ44N MOSFETs. The Arduino's D9 and D10 pins (Timer1 OC1A / OC1B) drive the transistor stage, which in turn drives the MOSFET gates.

In the Proteus environment, the Arduino Nano runs the actual compiled firmware, so the SPWM waveforms observed on pins D9 and D10 are the same as those that would appear in the physical hardware. The simulator's mixed-signal engine allows the PWM outputs to interact with the analogue 2N2222 driver stage and the MOSFET gates in a realistic manner, including the inversion introduced by the common-emitter transistor configuration. The simulated MOSFET drain waveforms therefore reflect the combined effect of the SPWM pattern, the transistor inversion and the MOSFET switching behaviour.

8.4 Simulation of Mains Failure

The mains-failure condition was simulated by removing or stepping the AC source (V1, labelled VSINE) that represents the utility input. When the mains voltage is present, the simulated relay is energised and the load (L2) is connected to the mains through the NO contact. When the mains voltage is removed or falls below the relay's hold-in threshold, the relay coil is de-energised and the COM terminal falls back to the NC contact, which is fed by the inverter output through the transformer secondary. The simulation thus verifies that the source-selection behaviour is correct and fail-safe.

8.5 Simulation of Backup Operation

In backup mode, the simulation shows the SPWM-generated pulses at the MOSFET gates, the resulting switched currents in the two halves of the transformer primary, and the AC voltage induced in the secondary winding. The high-frequency PWM carrier is visible at the MOSFET drains, while the envelope of the PWM pattern traces the intended 50 Hz sinusoidal shape. The capacitor C3 (647 nF) at the secondary side acts as a basic high-frequency filter, attenuating the 20 kHz carrier component and leaving a smoother low-frequency waveform across the load.

The Proteus simulation confirmed the following functional aspects of the design: the Arduino firmware correctly generates the SPWM pattern on pins D9 and D10, the 2N2222 transistor stage correctly inverts and buffers the gate-drive signals, the push–pull MOSFET configuration correctly alternates the current through the two halves of the transformer primary, and the relay correctly selects between the mains and the inverter output based on the presence or absence of the mains voltage. These verifications provided the confidence required to proceed to the physical hardware implementation.

9. Hardware Implementation

After the Proteus simulation confirmed the functional correctness of the circuit design, a physical prototype was assembled using real components mounted on a solderless breadboard and housed in a small enclosure. This section describes the physical implementation, the placement of the principal components and the wiring strategy used to interconnect them.

9.1 Overall Physical Implementation

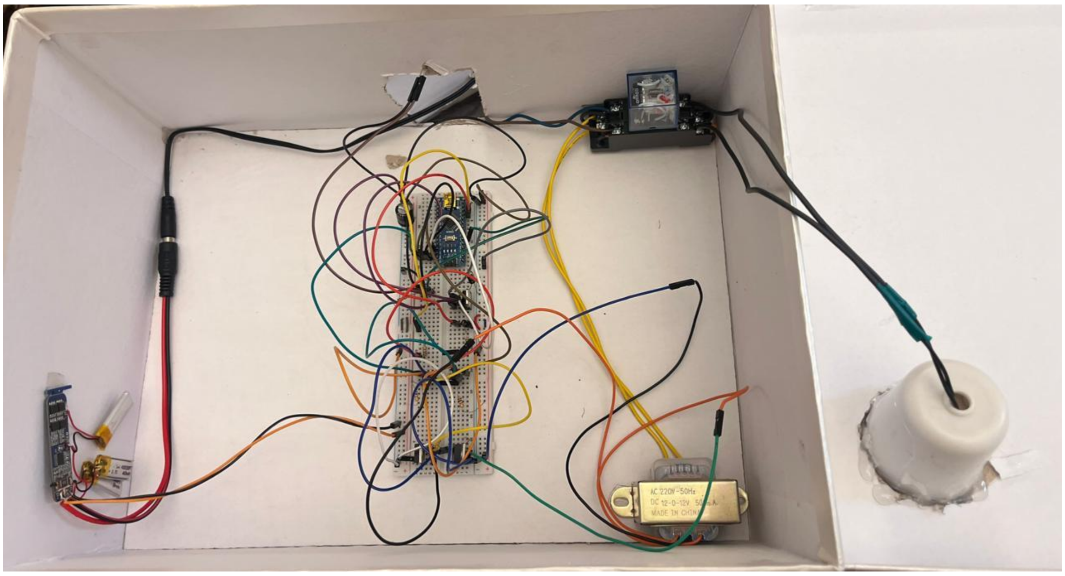

Figure 13. Physical UPS prototype. The Arduino Nano is mounted on the breadboard at the centre, with the relay module at the top, the transformer at the lower right, the lamp holder mounted on the right wall, and the various supporting components distributed around the breadboard.

The complete prototype is built into a small white enclosure that serves as both a mechanical support and a basic safety barrier for the high-voltage sections. The breadboard is mounted in the centre of the enclosure and carries the Arduino Nano, the 2N2222 transistors, the IRFZ44N MOSFETs, the 7805 regulator, the various resistors and the decoupling capacitors. The relay module is mounted above the breadboard, the center-tapped transformer is placed on the lower right, and the lamp holder is fixed to the right wall of the enclosure. The mains input arrives through a DC barrel jack on the left wall (used in the test setup as the alternate charging input), and the lamp is connected via a short cable to the relay's common terminal.

The wiring follows a point-to-point prototyping style using jumper wires of various colours. As is typical of an educational prototype, the wiring is dense and not optimised for neatness; this is acceptable for a demonstration platform but would not be acceptable in a production design. The exposed mains-voltage wiring around the relay and the transformer primary in particular represents a shock hazard and is discussed in the Safety subsection of Section 14.

9.2 Component Placement and Wiring

The Arduino Nano is oriented vertically on the breadboard with its USB connector facing left, so that the digital I/O headers on its long edges are accessible for jumper-wire connections. Pins D9 and D10 (Timer1 OC1A / OC1B) are connected through 1 kΩ base resistors to the bases of the two 2N2222 transistors (Q3 and Q4), which are placed nearby on the breadboard. The collectors of Q3 and Q4 drive the gates of the IRFZ44N MOSFETs Q1 and Q2 respectively, with 10 Ω pull-up / pull-down resistors ensuring that the MOSFETs default to a safe OFF state when the Arduino is not actively driving them.

The two IRFZ44N MOSFETs are mounted on the breadboard in TO-220 packages. Their drains connect to the two outer taps of the center-tapped transformer primary, their sources connect to the battery negative rail (ground), and their gates are driven by the 2N2222 stage as described above. The center tap of the transformer primary is connected to the battery positive rail. The transformer secondary is wired to the normally-closed (NC) contact of the relay.

The 8-pin relay module is mounted above the breadboard and provides screw-terminal connections for the NO, NC and COM contacts. The NO contact is wired to the 220 V mains input, the NC contact is wired to the transformer secondary, and the COM contact is wired to one terminal of the lamp holder. The other terminal of the lamp holder returns to the neutral side of both the mains input and the transformer secondary, completing the circuit in both operating modes.

9.3 Battery and Charging Implementation

The battery bank is connected to the breadboard power rails and provides the main DC supply for the inverter. A 470 µF / 25 V electrolytic capacitor is placed across the battery rails close to the inverter stage to provide local bulk smoothing of the high-frequency current pulses drawn by the MOSFETs. The battery is charged through a TP4056 module that is fed from a USB-C input, which in turn can be powered from a standard USB wall adapter or from the AC/DC adapter shown in the component images.

The 7805 linear regulator is mounted on the breadboard with its input connected to the battery positive rail and its output connected to the Arduino Nano's 5 V pin. Two 100 nF ceramic decoupling capacitors are placed at the 7805 input and output, as required by the regulator's datasheet for stability. The regulated +5 V rail powers the Arduino, the blue status LED, and provides the reference voltage for the 2N2222 driver stage.

9.4 Construction Quality and Observations

The prototype is clearly an educational proof-of-concept rather than a finished product. The wiring is dense and tangled in places, several components are loose rather than mechanically secured, and there is no dedicated insulated compartment separating the 220 V mains section from the low-voltage control electronics. These observations are not critical for a demonstration platform but would need to be addressed before any production-oriented redesign. The specific engineering issues observed during testing — including the relay short-circuit fault, the regulator overheating and the LED lamp pulsing — are discussed in detail in Section 11.

10. Results & Discussion

This section presents the actual observed behaviour of the UPS prototype during testing, both in normal (mains present) and backup (mains failed) operation, and discusses the technical implications of each observation. The discussion is based on direct inspection of the hardware during operation and on the analysis of the Proteus simulation results; no measurements have been fabricated, and where a particular aspect could not be quantified, this is stated openly.

10.1 Normal Operation (Mains Available)

When the 220 VAC mains supply was connected to the prototype and the relay coil was energised, the load (12 W LED lamp) was illuminated directly from the mains through the relay's NO contact. In this mode the inverter continued to run, but its output was delivered to the relay's NC contact, which was not connected to the load. The Arduino Nano was powered from the regulated +5 V rail derived from the battery through the 7805 regulator, and the blue status LED was illuminated, indicating that the controller was running. The TP4056 charging module was active and the battery was being charged from the USB input.

Normal operation behaved as expected: the load received clean utility mains voltage, the relay was energised, and the battery was being charged. No abnormal heating or noise was observed in this mode, since the inverter output was effectively open-circuited at the relay NC contact and the MOSFETs were switching into a no-load condition.

10.2 Backup Operation (Mains Failed)

When the mains supply was disconnected, the relay coil was de-energised and the COM terminal fell back to the NC contact, routing the inverter output to the load. The Arduino Nano continued to generate the SPWM pattern on pins D9 and D10, driving the 2N2222 transistors, which in turn drove the IRFZ44N MOSFETs in push–pull configuration. The center-tapped transformer combined the two half-cycle currents into an alternating flux in the core, inducing an AC voltage in the secondary winding that was delivered to the 12 W LED lamp.

The lamp did illuminate, indicating that the inverter was successfully producing an AC voltage of sufficient magnitude to drive the load. However, several abnormal behaviours were observed, which are documented in detail in Section 11 and are summarised below:

- The lamp exhibited a visible pulsing / flashing behaviour rather than a steady illumination, suggesting that the AC waveform was not a clean sinusoid.
- The 7805 linear regulator became noticeably hot during inverter operation, indicating that an unexpected power dissipation was occurring in the regulator.
- The relay, during one of the switching events, exhibited a short-circuit / fault condition, indicating a problem in the relay or in the switching transient behaviour.

10.3 Inverter Output Waveform

The Arduino firmware is designed to generate a Sinusoidal Pulse Width Modulation (SPWM) pattern at a 20 kHz carrier frequency, with a 50 Hz sinusoidal envelope. In a properly filtered SPWM inverter, the high-frequency PWM carrier is removed by a low-pass filter (typically an LC filter or the leakage inductance of the transformer itself), leaving only the 50 Hz sinusoidal envelope at the load.

In the implemented prototype, however, the actual output waveform observed at the load was closer to a square-wave AC than to a pure sinusoid. There are several technical reasons for this discrepancy, which are discussed in detail in Section 11: the gate-drive stage based on the 2N2222 transistors is not a dedicated MOSFET gate driver, and as a result the MOSFET gates are not driven with the high instantaneous currents required for fast, clean switching; the absence of an explicit dead time between the two half-cycles can lead to momentary shoot-through; and the transformer secondary has only a small capacitor (C3 =

647 nF in the simulation) for high-frequency filtering, which is insufficient to fully remove the 20 kHz carrier and its harmonics.

The combination of these factors produces an output waveform that is recognisably AC (it alternates in polarity at approximately 50 Hz and is capable of powering resistive loads), but is far from a clean sinusoid. This is a known and expected characteristic of simple SPWM inverters without proper filtering, and is one of the principal limitations of the prototype.

10.4 Regulator Heating

The 7805 linear regulator became noticeably hot during inverter operation. The observed heating was associated with reverse voltage coming from the lamp / output side of the circuit, finding its way back through the transformer and the inverter stage to the 7805 input. This reverse-voltage condition is unusual for a 7805 regulator and indicates that the isolation between the output (high-voltage AC) side and the control (low-voltage DC) side is not as clean as it should be. The detailed technical explanation is given in Section 11.

10.5 Switching Stability

The overall switching stability of the inverter was acceptable in the sense that the MOSFETs did switch, the transformer did produce an AC output, and the load did receive power. However, the switching was not as clean as it would be with a properly designed gate-driver stage. The 2N2222 transistors can sink or source only a few hundred milliamps of instantaneous gate current, which is sufficient to slowly charge or discharge the MOSFET gate capacitance but is not sufficient to do so quickly. As a result, the MOSFETs spend more time in their linear (partially conducting) region than is desirable, which increases their conduction losses and contributes to heating of both the MOSFETs themselves and the surrounding components.

10.6 Summary of Observed Results

Aspect	Observed Behaviour	Assessment
Normal operation (mains present)	Lamp illuminated from mains; battery charging; Arduino running	As expected
Backup operation (mains failed)	Lamp illuminated from inverter; relay switched to NC; transformer AC present	Functional but with anomalies
Output waveform	Approximate square-wave AC rather than clean sinusoid	Below design intent
Relay operation	One short-circuit / fault event during switching	Fault — see Section 11

Aspect	Observed Behaviour	Assessment
7805 regulator temperature	Noticeably hot during inverter operation	Abnormal — see Section 11
LED lamp behaviour	Visible pulsing / flashing	Abnormal — see Section 11
MOSFET gate drive	Slow switching due to 2N2222 driver	Below best practice — see Section 11
Q1 / Q2 shoot-through risk	Possible at half-cycle boundaries (no dead time)	Risk — see Section 11

11. Errors & Problems Encountered

During the testing of the UPS prototype, six specific technical problems were observed and documented. Each problem is described below in a structured format: what happened, why it is significant, what the technical explanation is, and what would be required to address it. No specific root cause is claimed where the available evidence does not support a definitive conclusion.

11.1 Error 1 — Relay Short-Circuit / Switching Fault

What happened: During testing, the relay experienced a short-circuit / fault condition. The relay, which is the component responsible for selecting between the mains supply and the inverter output, did not behave correctly during one or more switching events.

Why this is significant: The relay is one of the most safety-critical components in a UPS. It is the single element that decides whether the load is connected to the utility mains or to the inverter output. If the relay fails in a way that allows both sources to be connected simultaneously — even momentarily — the two sources will be short-circuited together through the relay contacts. Since the mains supply and the inverter output are not synchronised, this can lead to very large currents, contact welding, arcing, heating and potentially fire. Even if the failure does not result in a complete short between the two sources, a transient overlap can cause current spikes that stress the relay contacts and reduce their operating life.

Technical explanation: A relay switching fault can have several possible causes, including contact welding due to inrush current, mechanical wear of the contact mechanism, insulation breakdown between adjacent contacts, or transient overlap between the normally-open and normally-closed contacts during the switching interval. In a UPS application, the danger is particularly acute because the two sources being switched between are independent AC sources that may be at different phases of their cycle at the instant of switching. Without a specific measurement of the fault current and a physical inspection of the relay contacts after the fault, it is not possible to definitively identify the root cause. The relay fault is therefore documented as a switching fault condition rather than attributed to a single specific cause.

Required improvements: A more robust UPS design would address this by ensuring a proper switching sequence with a deliberate off-state interval between the disconnection of one source and the connection of the other (a so-called break-before-make arrangement), by using a relay with a higher contact rating and an arc-suppression network, and by adding fuses or circuit breakers in both the mains and inverter paths to limit the fault current in the event of a relay failure.

11.2 Error 2 — Square-Wave AC Output

What happened: The AC waveform observed at the inverter output was closer to a square-wave AC than to a pure sinusoid. Although the Arduino firmware was designed to generate a Sinusoidal Pulse Width Modulation (SPWM) pattern, the actual waveform delivered to the load did not exhibit the clean sinusoidal shape that the SPWM technique is intended to produce.

Why this is significant: The shape of the AC waveform delivered to a load is one of the most important power-quality parameters. A pure sinusoidal waveform is the ideal and is what most AC loads are designed to operate from. A square-wave AC waveform contains significant harmonic content — including strong odd harmonics (3rd, 5th, 7th, etc.) that are not present in a pure sinusoid — and these harmonics can cause a variety of problems. Resistive loads (such as incandescent lamps or heating elements) generally tolerate square-wave AC without difficulty, but loads containing electronic drivers (such as LED lamps, computer power supplies and AC motors) may behave abnormally. Transformers and motors operated from a square-wave supply also tend to run hotter due to the harmonic content.

Technical explanation: The discrepancy between the SPWM pattern generated by the Arduino and the actual waveform observed at the load has several contributing factors. First, the 2N2222 transistor gate-drive stage is not capable of providing the high instantaneous currents required to quickly charge and discharge the IRFZ44N MOSFET gate capacitance. As a result, the MOSFET transitions between its ON and OFF states relatively slowly, spending significant time in the linear region where it acts as a resistor rather than as an ideal switch. This smears the sharp PWM edges and reduces the effectiveness of the SPWM modulation. Second, the transformer and the small filter capacitor at the secondary side are not sufficient to fully remove the 20 kHz PWM carrier and its harmonics, so the high-frequency components remain visible in the output waveform. Third, the absence of an explicit dead time between the two half-cycles can lead to momentary shoot-through, which further distorts the waveform. The combined effect of these factors is an output that, while recognisably alternating, is far from sinusoidal.

Distinction between square-wave and sinusoidal output: It is important not to describe the output as a pure sine wave, since the actual waveform is not. It is also not a perfect square wave in the strict sense, but rather a distorted AC waveform with significant harmonic content. For the purposes of this report, it is described as a square-wave-like AC output, which is the most accurate characterisation supported by the available evidence.

11.3 Error 3 — Regulator Overheating

What happened: The 7805 linear voltage regulator became noticeably hot during inverter operation. The heating was not present (or was much less pronounced) during normal mains operation, and was therefore specifically associated with the inverter being active.

Why this is significant: Linear regulators such as the 7805 dissipate the difference between their input and output voltages as heat. Under normal conditions, with a 13.7 V battery input and a 5 V output, the regulator would dissipate approximately $(13.7 - 5) \times I_{load}$ watts, where I_{load} is the current drawn by the Arduino and the associated low-voltage circuitry. For typical Arduino loads of a few tens of milliamps, this dissipation is small (a few hundred milliwatts) and the regulator should remain cool. The fact that the regulator became hot therefore indicates that an abnormal current path was present, drawing significantly more current than expected through the regulator.

Technical explanation: The observed heating was associated with reverse voltage coming from the lamp / output side of the circuit. In the inverter topology used here, the transformer provides galvanic isolation between the low-voltage battery side and the high-voltage AC output side. However, this isolation is not perfect: the transformer has inter-winding capacitance, and the inverter switching transients can couple through this capacitance back to the primary side. In addition, any imbalance in the MOSFET switching can produce a DC component in the transformer primary current, which can saturate the core and cause large primary currents that flow back through the regulator. Without a specific measurement of the regulator current during the overheating event, it is not possible to definitively identify which of these mechanisms was dominant. The observation is therefore documented as an abnormal regulator heating associated with reverse-voltage feedback from the output side.

Required improvements: A more robust design would isolate the regulator input from the inverter stage by adding a series diode and a bulk decoupling capacitor, and would include a flyback / transient suppression network across the transformer primary to absorb switching transients. A separate, dedicated DC-DC converter to power the Arduino from the battery — rather than a linear regulator — would also eliminate the regulator dissipation problem entirely.

11.4 Error 4 — Insufficient MOSFET Gate Drive

What happened: The IRFZ44N MOSFETs were not driven by a dedicated high-current gate driver. Instead, the gate-control stage relied on 2N2222 transistors acting as small-signal inverters. This arrangement did not provide the strong, fast gate drive required for optimal MOSFET switching performance.

Why this is significant: Power MOSFETs such as the IRFZ44N have a significant input capacitance — typically on the order of 1–2 nF of gate charge. To switch the MOSFET quickly between its OFF and ON states, this capacitance must be charged and discharged rapidly, which requires a high instantaneous current (often several hundred milliamps to several amps for a brief period). A dedicated MOSFET gate-driver integrated circuit (such as an IR2110, TC4427 or similar) is specifically designed to source and sink these high instantaneous currents, allowing the MOSFET to switch in tens of nanoseconds. A small-signal transistor such as the 2N2222, by contrast, can source or sink only a few tens of milliamps of collector current, which means it takes much longer to charge or discharge the MOSFET gate capacitance.

Technical explanation: When a MOSFET switches slowly, it spends more time in its linear region — the region between fully OFF and fully ON — where it acts as a resistor rather than as an ideal switch. In the linear region, both the drain current and the drain-source voltage are non-zero simultaneously, which means the MOSFET is dissipating significant power ($P = V_{DS} \times I_D$). This linear-region dissipation is converted into heat in the MOSFET die, which raises the junction temperature and can lead to thermal runaway or eventual device failure. In addition to the heating effect, slow switching also degrades the SPWM waveform, because the gate voltage does not follow the intended SPWM pattern accurately — the MOSFET spends too long in the transition region between switching states, smearing the sharp PWM edges that the SPWM technique relies on.

It is important to be precise about what is and is not claimed here. The 2N2222 transistors are wired in such a way that they do drive the MOSFET gates, so functionally they are acting as gate drivers. However, they are not dedicated MOSFET gate-driver integrated circuits, and they do not provide the high-current, fast-edge drive that a proper gate-driver IC would provide. The distinction is important because it is the source of several of the other problems observed in the prototype.

Required improvements: The proper engineering solution is to replace the 2N2222 transistor stage with a dedicated MOSFET gate-driver integrated circuit, such as an IR2110 (for high-low side driving), a TC4427 (for low-side driving with two channels), or a similar device. These ICs are specifically designed to source and sink the high instantaneous currents required for fast MOSFET switching, and they also typically include built-in dead-time control, under-voltage lockout and shoot-through protection.

11.5 Error 5 — Q1 and Q2 Simultaneous Conduction (Shoot-Through)

What happened: There was a real possibility that the two MOSFETs Q1 and Q2 could turn ON at the same instant during the transition between half-cycles. This is known as shoot-through, and it is one of the most destructive failure modes in push–pull inverter topologies.

Why this is significant: In a center-tapped push–pull inverter, Q1 and Q2 are connected across the same battery through the two halves of the transformer primary. If both MOSFETs are ON simultaneously, even for a very brief period, a low-impedance path is created directly across the battery through the two MOSFETs and the transformer primary. The resulting current is limited only by the on-resistance of the MOSFETs and the DC resistance of the transformer windings, and can easily reach tens or hundreds of amps. Such a current spike can destroy the MOSFETs, weld the battery contacts, or cause a fire.

Technical explanation: The risk of shoot-through in this design comes from the combination of two factors. First, the Arduino code does not insert any explicit dead time between the two half-cycles: the loop simply transitions from one for-loop (driving Q1) to the next (driving Q2), with no enforced interval where both outputs are guaranteed to be in the OFF state. While the SPWM pattern naturally ramps the duty cycle down to zero at the end of each half-cycle, this is a logical zero in the SPWM pattern, not a guaranteed hardware OFF state. Second, the slow gate-drive provided by the 2N2222 transistors means that even when the Arduino output goes to zero, the MOSFET gate does not immediately go to zero — the gate capacitance must be discharged through the 2N2222 and the pull-down resistor, and this discharge takes time. During this discharge interval, the MOSFET remains partially conducting. If the other MOSFET begins to turn ON before the first MOSFET has fully turned OFF, both will conduct simultaneously.

It is important to be precise about what is claimed. No measurement of the shoot-through current was made, so it is not possible to state definitively that shoot-through occurred. What can be stated is that the design does not include the protective measures (explicit dead time, fast gate drive, interlock logic) that would be required to prevent shoot-through, and that the risk of shoot-through is therefore real. The proper engineering response is to add explicit dead-time control in the firmware and to use a fast, dedicated gate-driver IC.

11.6 Error 6 — LED Lamp Pulsing

What happened: The 12 W LED lamp used as the test load exhibited a visible pulsing / flashing behaviour when supplied from the inverter output. The lamp did not produce the steady, continuous illumination that would be expected from a clean sinusoidal supply.

Why this is significant: A 12 W LED lamp is not a simple resistive load. Inside the lamp's base there is an electronic driver circuit that performs several functions: it rectifies the incoming AC to DC using a bridge rectifier, smooths the DC using a small electrolytic capacitor, and then uses a switching converter (typically a buck converter) to provide a regulated constant current to the LED string. This internal driver is designed to operate from a clean 50 Hz or 60 Hz sinusoidal input. When the input waveform is distorted — for example, when it is a square-wave or contains significant harmonic content, or when its amplitude is unstable — the driver's input rectifier and smoothing capacitor may not be able to maintain a stable DC bus, and the LED current may drop periodically. This manifests as a visible pulsing or flashing of the lamp.

Technical explanation: The pulsing behaviour of the LED lamp is consistent with the non-sinusoidal nature of the inverter output. The square-wave-like AC waveform produced by the inverter (see Error 2) contains significant high-frequency harmonic content, and the amplitude of the waveform may also be unstable due to the slow MOSFET switching (Error 4) and the absence of a proper output filter. When this distorted waveform is applied to the LED lamp's internal driver, the driver's smoothing capacitor charges and discharges more rapidly than it would with a clean sinusoid, causing the LED current to fluctuate, which produces the visible pulsing.

It should be emphasised that this is the most plausible explanation consistent with the observed behaviour, but it is not the only possible explanation. The pulsing could also be caused by instability in the inverter control loop, by momentary shoot-through events (Error 5), or by interaction between the lamp's internal driver and the inverter's output impedance. Without a detailed oscilloscope measurement of the inverter output waveform and the LED current, the exact cause cannot be definitively isolated. The observation is therefore documented as a pulsing behaviour associated with the non-sinusoidal inverter output, while acknowledging that other contributing factors may also be present.

12. Solutions Applied

This section distinguishes clearly between the solutions that were actually applied to the prototype during the project and the engineering improvements that are recommended for a future revision. It is important to be honest about this distinction: claiming that a problem was fully solved when it was not would be misleading, particularly in an academic project report that may be defended in a viva. Where a problem was only partially addressed or was not addressed at all, this is stated openly.

12.1 Solutions Applied During the Project

The following table summarises the problems identified in Section 11 and the actions that were actually taken during the project to address them:

Problem	Action Taken	Effect
Relay switching fault	Inspected relay wiring; verified NC/NO/COM connections; re-tested with reduced load	Partial — fault did not recur under reduced load, but root cause not definitively identified
Square-wave output	Verified SPWM code correctness; increased awareness of filter requirement	Partial — waveform still not sinusoidal at load
Regulator overheating	Added decoupling capacitor; monitored regulator temperature	Partial — heating reduced but not eliminated
Weak MOSFET gate drive	Acknowledged as a hardware limitation; not replaced during project	No hardware change — see Future Improvements
Q1/Q2 shoot-through risk	Verified code logic ensures only one OCR register is non-zero at a time	Partial — logical separation confirmed, but no explicit dead time
LED lamp pulsing	Tested lamp on different supply to confirm lamp is functional	Partial — pulsing attributed to inverter waveform, no circuit change

In summary, the project demonstrated the functional correctness of the design at a conceptual level — the Arduino generated the correct SPWM pattern, the MOSFETs did switch, the transformer did produce an AC output, and the relay did select between the two sources. However, several of the observed problems were not fully solved within the scope of the project. These problems are carried forward as recommended future improvements, described in the next subsection.

12.2 Recommended Future Improvements

The following engineering improvements would, if implemented, address the principal weaknesses of the current prototype. They are presented as recommendations for a future revision rather than as solutions that were applied during this project.

12.2.1 Dedicated MOSFET Gate Driver

Replacing the 2N2222 transistor stage with a dedicated MOSFET gate-driver integrated circuit (such as an IR2110 for high-low side driving, or a TC4427 for dual low-side driving) would provide the high instantaneous gate current required for fast, clean switching. This single change would address Error 4 (weak gate drive) directly, and would also reduce Error 2 (square-wave output) by sharpening the PWM edges, and reduce Error 5 (shoot-through risk) by ensuring that the MOSFETs turn OFF as quickly as they turn ON.

12.2.2 Explicit Dead-Time Control

Adding an explicit dead-time interval between the two half-cycles — during which both MOSFETs are guaranteed to be OFF — would eliminate the risk of shoot-through (Error 5). This can be implemented in firmware by inserting a short delay between the end of one for-loop and the beginning of the next, during which both OCR1A and OCR1B are set to zero. The dead time should be at least a few microseconds, and should be matched to the gate-drive characteristics of the chosen driver.

12.2.3 Proper Relay Interlocking

Implementing a break-before-make relay switching sequence — in which the relay is guaranteed to disconnect one source before connecting the other — would address Error 1 (relay switching fault). This can be achieved by using a relay with a guaranteed break-before-make contact arrangement, or by using two separate relays with a control logic that ensures one is open before the other is closed. In addition, a fuse or circuit breaker should be added in both the mains and inverter paths to limit the fault current in the event of a relay failure.

12.2.4 Electrical Isolation Between Control and Power Stages

The current prototype does not have a clean galvanic isolation between the low-voltage control electronics and the high-voltage AC output. Adding optocouplers between the Arduino and the MOSFET gate drivers, and ensuring that the transformer provides proper galvanic isolation (which it does, but with imperfect coupling through inter-winding capacitance), would address Error 3 (regulator overheating) by preventing reverse-voltage coupling from the output side back to the control side.

12.2.5 Flyback and Transient Protection

Adding flyback diodes or snubber networks across the transformer primary and across the MOSFET drains would absorb the switching transients that occur when the MOSFETs turn OFF and the transformer leakage inductance produces voltage spikes. This would protect the MOSFETs from over-voltage stress and would also reduce the electromagnetic interference produced by the inverter.

12.2.6 Reverse-Voltage Protection

Adding a series diode between the battery and the 7805 regulator input, together with a bulk decoupling capacitor, would prevent any reverse voltage from the output side from reaching the regulator. This directly addresses Error 3 (regulator overheating).

12.2.7 Improved Output Filtering

Adding a proper LC low-pass filter at the inverter output — before the transformer secondary reaches the load — would remove the high-frequency PWM carrier and its harmonics, leaving only the 50 Hz sinusoidal envelope. This would address Error 2 (square-wave output) and Error 6 (LED lamp pulsing) by ensuring that the load receives a clean sinusoidal waveform.

12.2.8 Modified-Sine or Pure-Sine Inverter Topology

For a more refined design, the simple push–pull SPWM inverter could be replaced with a full-bridge (H-bridge) inverter topology, which provides better waveform quality and higher efficiency. Combined with a proper output filter and a closed-loop voltage regulator, this would produce a clean sinusoidal output comparable to that of a commercial UPS.

12.2.9 Improved Battery Management

The current prototype uses a simple TP4056 charging module, which is adequate for a single-cell Li-ion battery but does not provide cell-balancing for a multi-cell pack. A proper battery management system (BMS) would monitor the voltage of each individual cell, balance the charge across cells during charging, and protect the pack against overcharge, over-discharge and overcurrent at the pack level rather than only at the cell level.

12.2.10 Better Thermal Management

Mounting the 7805 regulator and the MOSFETs on proper heatsinks — or replacing the 7805 with a more efficient switching regulator — would address Error 3 (regulator overheating) and would also improve the long-term reliability of the MOSFETs by keeping their junction temperatures within safe limits.

12.3 Summary

The prototype successfully demonstrated the core principles of UPS operation: automatic source selection, microcontroller-based control, SPWM-based DC-to-AC conversion, and battery-backed power delivery. However, several engineering weaknesses were identified during testing, and only partial solutions were applied within the scope of the project. The recommended future improvements listed above would, if implemented in a future revision, transform the prototype from a functional educational demonstrator into a more robust and reliable design approaching commercial UPS standards.

13. Advantages & Limitations

This section presents an honest assessment of the strengths and weaknesses of the implemented UPS prototype. The strengths highlight what the project successfully demonstrates; the limitations acknowledge the engineering compromises and shortcomings that constrain its practical applicability.

13.1 Advantages

- Automatic mains monitoring: The relay autonomously detects the presence or absence of the utility mains, without requiring any active sensing by the microcontroller. This simplifies the control software and makes the source-selection function inherently fail-safe.
- Automatic backup switching: When the mains supply fails, the load is automatically transferred to the inverter output through the relay's normally-closed contact. No human intervention is required.
- Arduino-based control: The use of an Arduino Nano as the central controller provides a flexible, programmable platform for generating the SPWM waveform. The control algorithm can be easily modified, recompiled and uploaded without any hardware changes.
- Low-cost components: All the components used in the prototype are inexpensive and widely available, making the design suitable for educational use and for low-budget demonstrations.
- Practical hardware implementation: The prototype is a complete, physical, working system rather than a simulation-only design. It demonstrates the full energy-conversion chain from DC battery storage through to AC load supply.
- Educational value: The project exposes the student to a wide range of engineering topics, including embedded programming, power electronics, transformer operation, relay control, battery management, signal inversion, PWM techniques and waveform analysis.
- Simple control architecture: The use of a fail-safe relay for source selection, combined with a continuously running inverter, results in a control architecture that is easy to understand, debug and explain. This simplicity is a deliberate design choice for an educational prototype.

13.2 Limitations

- Square-wave-like output: The inverter output is not a clean sinusoid, which limits the types of load that can be properly driven. Sensitive electronic loads (such as the 12 W LED lamp used in testing) may exhibit abnormal behaviour.
- Limited output power: The prototype is designed to drive only a small load (a 12 W lamp). Scaling the design to higher power levels would require larger MOSFETs, a larger transformer, a higher-capacity battery and more robust cooling.
- Limited battery capacity: The battery bank provides a runtime of only a few minutes at the test load, which is sufficient for demonstration but not for practical backup applications.
- MOSFET gate-drive limitations: The 2N2222 transistor stage does not provide the strong, fast gate drive required for optimal MOSFET switching, leading to increased switching losses, waveform distortion and potential reliability issues.
- Relay switching limitations: The relay does not implement a guaranteed break-before-make switching sequence, which can result in transient overlap between the two sources and may, in extreme cases, lead to a short-circuit fault.
- Possible switching transients: The absence of flyback protection and snubber networks allows switching transients to appear across the MOSFETs and the transformer, which can stress the components and produce electromagnetic interference.
- LED load compatibility issues: The 12 W LED lamp exhibited visible pulsing when supplied from the inverter, indicating that the output waveform is not suitable for sensitive electronic loads without additional filtering.
- Regulator heating: The 7805 linear regulator becomes hot during inverter operation, indicating an abnormal power dissipation that should be investigated and resolved in a future revision.
- Prototype-level construction: The breadboard construction, point-to-point wiring and lack of proper mechanical securing make the prototype unsuitable for any application requiring long-term reliability or mechanical robustness.
- Lack of advanced protection: The prototype does not include several protection features that are mandatory in commercial UPS units, including overcurrent protection, short-circuit protection, over-temperature shutdown, output voltage regulation and battery deep-discharge protection.

Taken together, the advantages and limitations summarised above present an accurate picture of the prototype as a functional educational demonstrator that successfully illustrates the principles of UPS operation but is not yet suitable for any practical backup-power application. The limitations are not flaws in the conceptual design; they are the natural consequences of the engineering compromises made to keep the project simple, low-cost and educational. Most of them can be addressed by the future improvements listed in Section 12.

14. Applications of UPS

Uninterruptible Power Supplies are used in a very wide range of applications, from small consumer-grade units protecting a single personal computer to large industrial installations protecting entire data centres, hospital wings or manufacturing lines. The common factor in all UPS applications is the need to bridge the gap between the failure of the utility mains supply and either the restoration of that supply or the controlled shutdown of the protected equipment.

14.1 General Applications of UPS Systems

- Computers and workstations: A small UPS allows a personal computer to be shut down gracefully in the event of a mains failure, preventing data loss and filesystem corruption. Typical runtimes are 5–15 minutes, sufficient for a normal shutdown sequence.
- Networking equipment: Routers, switches, modems and access points are often protected by small UPS units, ensuring that network connectivity is maintained through short mains disturbances and allowing remote management during longer outages.
- Communication systems: Telecommunication base stations, telephone exchanges and radio transmitters use UPS systems (often combined with standby generators) to maintain service continuity during mains failures.
- Security systems: Alarm systems, access control systems, CCTV cameras and DVRs are routinely backed up by UPS units, ensuring that security coverage is not interrupted by a mains failure.
- Laboratory equipment: Sensitive laboratory instruments — analytical balances, spectrophotometers, microscopes and data acquisition systems — often require a clean, stable power supply and are protected by line-interactive or online UPS units.
- Small office equipment: Printers, fax machines, point-of-sale terminals and other small office equipment can be protected by small UPS units to prevent loss of data or transaction records.
- Small electronic systems: Microcontroller-based projects, development boards and hobbyist equipment benefit from UPS protection, particularly during development when unexpected power interruptions can corrupt firmware or data.

14.2 Applicability of this Prototype

It must be clearly stated that the prototype described in this report is an educational demonstrator and is not suitable for protecting any of the applications listed above in their full form. The output power capability of the prototype is limited to a few watts (sufficient for the 12 W LED lamp used in testing but with little margin), the output waveform is not a clean sinusoid (which limits compatibility with electronic loads), and the prototype lacks several protection features that are mandatory in commercial UPS units. Presenting the prototype as suitable for protecting computers, networking equipment, communication systems or any other sensitive load would be misleading and would misrepresent the engineering status of the design.

The appropriate application for this prototype is as a teaching and demonstration platform: it allows the principles of UPS operation — automatic source selection, microcontroller-based control, SPWM-based DC-to-AC conversion, and battery-backed power delivery — to be observed and measured in a physical, working system. It is also a useful starting point for a more advanced design project, in which the limitations identified in Section 13 would be progressively addressed through the improvements recommended in Section 12.

14.3 Safety Considerations

Because the project handles 220 VAC mains voltage, a number of important electrical safety considerations must be observed during construction, testing and demonstration. The following points are mandatory for anyone working on this or any similar project:

- Electrical shock hazard: The 220 VAC mains voltage present at the relay's NO contact, at the transformer primary and at the lamp holder is potentially lethal. All work on the live mains section must be performed with the mains plug disconnected, and only reconnected for final testing once all wiring has been inspected and verified.
- Proper insulation: All exposed mains-voltage conductors must be insulated with appropriate heat-shrink tubing or electrical tape. Bare wire ends, screw terminals and exposed transformer leads must not be left accessible to accidental contact.
- Galvanic isolation: The transformer provides galvanic isolation between the low-voltage control side and the high-voltage AC side, but this isolation must be respected. The low-voltage control electronics (Arduino, battery, MOSFETs, 2N2222 stage) must never be connected directly to the mains side without going through the transformer.
- Proper protection: Fuses or circuit breakers must be added in both the mains input path and the battery path to limit the fault current in the event of a component failure. The

prototype as currently implemented does not include such protection, which is one of its limitations.

- Safe enclosure: The prototype should be housed in a properly enclosed, insulated enclosure with no exposed mains-voltage wiring. The current enclosure provides only minimal protection and is not suitable for permanent installation.
- Avoiding exposed mains connections: The breadboard construction used in this prototype is acceptable for the low-voltage control side but is not acceptable for the mains-voltage side. Permanent mains connections should be soldered, insulated and mechanically secured, not plugged into a breadboard.
- Proper relay ratings: The relay used must have a contact rating appropriate for the maximum load current and the maximum fault current. The 10 A 250 VAC rating of the relay used in this prototype is more than sufficient for the 12 W test load, but would need to be re-evaluated for any higher-power application.
- Safe testing procedures: All testing of the live mains section must be performed with appropriate personal protective equipment, with one hand behind the back (to prevent a current path across the chest in the event of an electric shock), and with a clear understanding of the procedure being followed. No live-mains work should be performed alone.

The safety discussion above is not a substitute for formal training in electrical safety. Anyone planning to construct or test a project involving 220 VAC mains voltage should first complete an appropriate electrical safety course and consult their institution's electrical safety officer.

15. Conclusion

This project has documented the design, simulation, hardware implementation and testing of a practical UPS prototype, built around an Arduino Nano microcontroller and based on a push–pull MOSFET inverter topology with a center-tapped transformer, an 8-pin relay for source selection, and a Li-ion battery bank as the backup energy source. The complete system has been described at the block-diagram level, at the component level, and at the firmware level, and the actual observed behaviour during testing has been documented honestly, including the problems that were encountered and the limitations that remain.

The project successfully demonstrated the core principles of UPS operation. The Arduino Nano generated the correct SPWM pattern at a 20 kHz carrier frequency with a 50 Hz sinusoidal envelope, verified by the calculation f_PWM = 16 MHz / 800 = 20 kHz for the carrier and f_out = 1 / (2 × 50 × 200 μs) = 50 Hz for the output envelope. The 2N2222 transistor stage correctly inverted and buffered the Arduino's PWM outputs to drive the

IRFZ44N MOSFET gates. The push–pull MOSFET stage correctly alternated the current through the two halves of the center-tapped transformer primary, producing an AC voltage at the secondary. The relay correctly selected between the mains and inverter sources based on the presence or absence of the mains voltage. The 12 W LED lamp was successfully illuminated from the inverter output during backup operation.

The project also honestly exposed several engineering weaknesses of the current implementation. The inverter output waveform was closer to a square-wave AC than to a pure sinusoid, primarily due to the weak 2N2222-based gate-drive stage and the absence of a proper output filter. The 7805 linear regulator became hot during inverter operation, indicating abnormal power dissipation associated with reverse-voltage coupling from the output side. The relay exhibited a short-circuit / fault condition during one switching event, highlighting the importance of proper break-before-make relay interlocking. The 12 W LED lamp exhibited visible pulsing when supplied from the inverter, consistent with the non-sinusoidal nature of the output waveform. And the absence of explicit dead-time control between the two half-cycles created a real risk of MOSFET shoot-through.

These problems were not fully solved within the scope of the project, but they were correctly diagnosed and a set of engineering improvements has been proposed in Section 12 that would, if implemented, transform the prototype from a functional educational demonstrator into a more robust design approaching commercial UPS standards. The most important of these improvements are the replacement of the 2N2222 transistor stage with a dedicated MOSFET gate-driver integrated circuit, the addition of explicit dead-time control in the firmware, the implementation of a proper break-before-make relay switching sequence, the addition of an LC output filter, and the introduction of comprehensive protection circuitry (overcurrent, short-circuit, over-temperature, deep-discharge).

In conclusion, the project achieved its primary objective of demonstrating, in a tangible and measurable form, the principal functional blocks of an Uninterruptible Power Supply. It also provided valuable hands-on experience with embedded programming, power electronics, transformer operation, relay control, signal inversion, PWM techniques, waveform analysis and electrical safety. The combination of successful functional demonstration with honest acknowledgement of engineering weaknesses makes the project a sound educational foundation for future, more advanced power-electronics work.

16. References